Chaitanya Mittal

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Technical Skills

Languages: C/C++, Java, Python, JavaScript, TypeScript, SQL
Systems & Backend: Linux, Spring Boot, Express.js, Flask, REST APIs, MVC Architecture
Databases: PostgreSQL, SQLite, Supabase, Prisma
DevOps & Testing: Git, GitHub Actions, CI/CD, Docker, AWS Elastic Beanstalk, Google Cloud Platform (GCP), Vercel
Frontend: React.js, Next.js, Tailwind CSS, Socket.IO, HTML/CSS
Software Practices: Object-Oriented Programming, Design Patterns, Debugging, Integration Testing, Agile/Scrum

Technical Experience

Full Stack Developer
SFU Blueprint (Richmond Poverty Reduction Coalition) **Sep 2025 – Present**

- Engineered a 3-step signup flow using **React** and **Supabase Auth**, implementing email/password authentication, multi-field profile collection, and email verification for front-end onboarding of member and admin users.
- Built a shared **React** component library (**shadcn/ui**, **Tailwind CSS**) applying reusable design patterns with built-in validation and error handling, reused across 5+ form screens including signup, renewal, and profile editing flows.
- Implemented admin-only user deletion via **Supabase RPC** and **auth.admin.deleteUser**, enforcing access control across both database and authentication layers — delivered within an Agile sprint cycle with code review and team QA.

Technical Projects

CrowdSurf | *Github* **React | Express | PostgreSQL | Prisma | Socket.IO | Docker**

- Built and deployed a full-stack event photo-sharing platform using **React** and **Express**, designing a 12-entity **PostgreSQL** schema in **Prisma** and a **REST API** of 30+ endpoints with **Joi**-based request validation.
- Implemented **JWT** authentication using an asymmetric ES256 keypair with refresh-token sessions, architecting a two-layer **role-based authorization** system enforced via Express middleware to block unauthorized API access.
- Integrated **Socket.IO** to broadcast photo uploads and likes in real time across event rooms, wrapping multi-table writes in transaction blocks to guarantee atomicity for operations like event creation and join-request approval.
- Wrote 46 integration tests across 23 suites using **Supertest** against a live **PostgreSQL** instance, and containerized the backend with a multi-stage **Docker** build deployed via **GCP** backend and **nginx** frontend.

SFU Course Planner | *Live Demo* | *Github* **Spring Boot | Next.js | AWS | OAuth | JWT | PostgreSQL**

- Built and deployed a full-stack web application using **Next.js** and **Spring Boot**, implementing **MVC architecture** and 30+ **REST API endpoints** to help 35+ users analyze enrollment trends across 500+ courses.
- Diagnosed and resolved a performance bottleneck causing 13–15s response times by redesigning data retrieval and persisting course and offering metadata in **PostgreSQL**, reducing end-user latency to under 5s.
- Developed an administrative dashboard for user management and system monitoring, implementing scheduled email notifications with **Spring Scheduler** and **Resend API** to automate status updates.
- Designed and deployed a cloud architecture using **Vercel**, **AWS Elastic Beanstalk**, and **Cloudflare** for secure HTTPS communication between frontend and backend services. ?¿

EkantikSearch | *Live Demo* | *Github* **React | TypeScript | Flask | SQLite | CI/CD**

- Built and deployed a full-stack web application using **React.js** and **Flask**, engineering a **TypeScript** frontend that maps multilingual keyword queries to exact YouTube timestamps across 800+ indexed videos.
- Engineered an automated data integration pipeline using **GitHub Actions** and **Flask** scripts to process **YouTube Data API** content with incremental updates, snapshot backups, and failure-alert monitoring workflows.
- Integrated the **Azure Translator API** for Hindi-to-English transliteration and optimized **SQLite** query performance using structured indexing, reducing multilingual search retrieval latency by 40%.

Education

SFU: Bachelor of Computer Science — GPA: 3.81 **Expected Graduation: Dec 2028**

∗ **Dean's Honour Roll:** Summer 2024, Fall 2024, Spring 2025, Summer 2025
∗ Relevant Courses: Web Development II (CMPT 372), Software Engineering (CMPT 276), Object-Oriented Design (CMPT 213), Data Structures & Algorithms (CMPT 225)